

3 Coordinate geometry in the (x,y) plane	3.1	Understand and use the equation of a straight line, including the forms $y - y_1 = m(x - x_1)$ and $ax + by + c = 0$; Gradient conditions for two straight lines to be parallel or perpendicular. Be able to use straight line models in a variety of contexts.	To include the equation of a line through two given points, and the equation of a line parallel (or perpendicular) to a given line through a given point. $m' = m$ for parallel lines and $m' = -\frac{1}{m}$ for perpendicular lines For example, the line for converting degrees Celsius to degrees Fahrenheit, distance against time for constant speed, etc.
	3.2	Understand and use the coordinate geometry of the circle including using the equation of a circle in the form $(x - a)^2 + (y - b)^2 = r^2$ Completing the square to find the centre and radius of a circle; use of the following properties: • the angle in a semicircle is a right angle • the perpendicular from the centre to a chord bisects the chord • the radius of a circle at a given point on its circumference is perpendicular to the tangent to the circle at that point.	Students should be able to find the radius and the coordinates of the centre of the circle given the equation of the circle, and vice versa. Students should also be familiar with the equation $x^2 + y^2 + 2fx + 2gy + c = 0$ Students should be able to find the equation of a circumcircle of a triangle with given vertices using these properties. Students should be able to find the equation of a tangent at a specified point, using the perpendicular property of tangent and radius.

Topics	What students need to learn:		
	Content		Guidance
3 Coordinate geometry in the (x, y) plane <i>continued</i>	3.3	Understand and use the parametric equations of curves and conversion between Cartesian and parametric forms.	For example: $x = 3\cos t$, $y = 3\sin t$ describes a circle centre O radius 3 $x = 2 + 5\cos t$, $y = -4 + 5\sin t$ describes a circle centre $(2, -4)$ with radius 5 $x = 5t$, $y = \frac{5}{t}$ describes the curve $xy = 25$ (or $y = \frac{25}{x}$) $x = 5t$, $y = 3t^2$ describes the quadratic curve $25y = 3x^2$ and other familiar curves covered in the specification. Students should pay particular attention to the domain of the parameter t, as a specific section of a curve may be described.
	3.4	Use parametric equations in modelling in a variety of contexts.	A shape may be modelled using parametric equations or students may be asked to find parametric equations for a motion. For example, an object moves with constant velocity from $(1, 8)$ at $t = 0$ to $(6, 20)$ at $t = 5$. This may also be tested in Paper 3, section 7 (kinematics).